Beyond Optimus 5-Prime: Reproduction and a CNN–Attention Hybrid for 5′UTR Translation Control

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Abstract

The 5′ untranslated region (5′UTR) of a messenger RNA controls how efficiently that message is translated, which makes it a central design variable for mRNA therapeutics. Sample et al. (2019) introduced *Optimus 5-Prime*, a convolutional network that predicts mean ribosome load (MRL) from a 50 nt 5′UTR and explains 93% of the variance of a massively parallel reporter assay. We reproduce that result from scratch in PyTorch (Paszke et al., 2019) ($r^2 = 0.919 \pm 0.006$ over three seeds), first validating the whole pipeline on a *synthetic* dataset whose generative rules we control, and we benchmark the model against three alternatives under one shared protocol. We then address one limitation of the original work: it evaluates a single architecture at a single data size, so it is not known whether the convolutional inductive bias is the right one for this task or merely a sufficient one. We run a controlled architecture × data-regime study at matched parameter counts and matched tuning budgets, and find that a CNN+attention hybrid reaches $r^2 = 0.942 \pm 0.001$ — clearly above the published CNN and the strongest model in our comparison — while a pure transformer on the same input fails outright in low-data regimes. In-silico mutagenesis shows that the hybrid refines the grammar the CNN already learns rather than learning a different one. Code, trained weights and all per-run metrics: https://github.com/anikag06/CHE_629.

1. Introduction

Translation of a eukaryotic mRNA begins when the small ribosomal subunit binds near the 5′ cap and scans along the transcript until it recognises a start codon. Everything the scanning ribosome encounters before the main coding sequence (CDS) therefore shapes how much protein the message yields: upstream AUG codons (uAUGs) that capture scanning ribosomes, upstream open reading frames (uORFs) that terminate before the CDS, the Kozak sequence context around each start codon, and secondary structure that impedes scanning (Kozak, 1986). The 5′UTR is thus a compact, highly interpretable regulatory element, and one that can be engineered: tuning a 50 nt sequence is a practical lever for mRNA vaccine and therapeutic design.

Sample et al. (2019) made this problem tractable for machine learning. They built a massively parallel reporter assay (MPRA) over a library of 280,000 random 50 nt 5′UTRs, measured the *mean ribosome load* (MRL) of each by polysome profiling, and trained a three-layer convolutional network, *Optimus 5-Prime*, to predict MRL from one-hot encoded sequence. The model explains 93% of held-out variance and generalises to human 5′UTRs and to clinical variants.

Contributions.

1. A faithful PyTorch reproduction ($r^2 = 0.919 \pm 0.006$ vs. 0.93 published), verified by loading the authors' own Keras weights into our implementation ($r^2 = 0.934$).
2. A synthetic 5′UTR simulator with known rules and a computable noise ceiling, used to validate the pipeline and to test *which* rules each model actually learns.
3. A three-benchmark comparison under one protocol, reporting regression and classification metrics (accuracy, macro precision/recall/F1, confusion matrices).
4. One focused extension: a controlled architecture × data-regime study, at matched parameter counts and matched tuning budgets, yielding a CNN+attention hybrid that improves on the published architecture ($r^2 = 0.942 \pm 0.001$ against 0.922 ± 0.002 for the CNN under the identical protocol), together with an in-silico mutagenesis check that the hybrid refines rather than replaces the grammar the CNN learns.

2. The Original Work

Biological problem. Given a 50 nt 5′UTR immediately upstream of a fixed reporter CDS, predict how heavily ribosomes load onto that transcript. MRL is a scalar summary

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of the polysome distribution: transcripts are fractionated by the number of bound ribosomes, and $\mathrm{MRL} = \sum_k k\, p_k$, where p_k is the fraction of that UTR's reads in polysome fraction k. High MRL means efficient translation initiation.

Dataset. A library of random 50-mers was cloned upstream of eGFP, transcribed *in vitro*, transfected into HEK293T cells, and separated over a sucrose gradient into 14 fractions, each sequenced. After filtering, the 280,000 most deeply sequenced UTRs were retained. The authors also produced the same library with uridine replaced by pseudouridine (Ψ) or N1-methylpseudouridine (m1Ψ) — the modified nucleotides used in mRNA vaccines (Karikó et al., 2008) — an mCherry-CDS library, a designed library containing 35,212 human 5′UTRs and 3,577 ClinVar variants, and a variable-length (25–100 nt) library.

Deep learning formulation. Each UTR is one-hot encoded to a 50×4 matrix. Optimus 5-Prime applies three 1-D convolutions (120 filters, width 8, `same` padding, ReLU), flattens, and passes through a 40-unit dense layer with 20% dropout to a single linear output. It is trained with MSE on standardised MRL using Adam (10^{-3}), batch 128, for three epochs — the authors stop there because the model begins to overfit. Performance is reported as r^2, the *squared Pearson correlation* between observed and predicted MRL on the 20,000 most deeply sequenced UTRs, which are held out.

3. Data and Experimental Setup

3.1. Real data and splits

We use GEO accession GSE114002. Following the original protocol exactly, UTRs are sorted by total read count, the top 280,000 are kept, the **top 20,000 form the test set**, and the remaining **260,000 form the training pool**. Because deeper sequencing means less measurement noise, this makes the test set the cleanest part of the data; we therefore also report a random-split control (Section 6.1).

The published recipe has no validation set: it fixes three epochs. For every experiment beyond the literal reproduction we carve a validation split from the training pool (10% at full data, 20% in small-data regimes) and use it for *all* model selection — learning rate, dropout, and early stopping. The 20,000 test UTRs are never used for any choice, and training subsets are drawn only from the training pool. Subset indices and seeds are saved to disk.

Noise ceiling. Because two biological replicates exist per chemistry, the best r^2 any model can achieve against replicate 1 equals the Pearson correlation between replicates: 0.971 (U), 0.916 (Ψ), 0.865 (m1Ψ). Every experiment in this report uses the unmodified (U) library, so 0.971 bounds every number we report on real data.

3.2. Synthetic dataset

The paper's library is often called "synthetic", but only the *sequences* are designed; the MRL labels are real measurements with real noise. To validate our pipeline before using real data, we built a genuinely synthetic dataset of 280,000 random 50-mers with labels generated from rules we control, following a leaky-scanning model:

$$\mathrm{MRL}_{\text{true}} = 1 + 7 \cdot e^{0.025\,\mathrm{MFE}} \prod_{i \in \mathcal{A}} (1 - r_i), \qquad (1)$$

where $\mathcal{A}$ is the set of upstream AUGs and $r_i = b_i \cdot \kappa_i$. The base repression b_i depends on what the uAUG does: 0.40 if an in-frame stop precedes the CDS (a self-contained uORF, permitting reinitiation), 0.75 if it is out of frame with no stop (an overlapping ORF), and 0.12 if it reads in frame into the CDS (an N-terminal extension). The Kozak factor $\kappa_i = 0.35 + 0.35\,[-3 \in \{A, G\}] + 0.30\,[+4 = G]$. Minimum free energy (MFE) is computed with ViennaRNA (Lorenz et al., 2011). Observed labels add $\mathcal{N}(0, 0.45^2)$, which fixes a computable noise ceiling of $r^2 = 0.918$.

This dataset serves three purposes: it validates preprocessing and training code against a known answer; it bounds achievable performance; and, because we know the planted rules, it lets us ask *which* rules a model has learned rather than only how well it scores.

4. Reproduction

Pipeline validation. Before training anything, we loaded the authors' released Keras weights (`main_MRL_model.hdf5`) into our PyTorch module and evaluated on our preprocessed test set, obtaining $r^2 = 0.934$. This confirms that our data processing, one-hot encoding, and architecture match theirs exactly. One subtlety: Keras `same` padding for an even kernel ($k = 8$) pads 3 on the left and 4 on the right, which must be replicated for the weights to transfer.

Synthetic first, then real. On the synthetic dataset the CNN reaches $r^2 = 0.856 \pm 0.006$ against a noise ceiling of 0.918. Crucially, in-silico insertion of a single ATG into uAUG-free sequences shows the model recovers the planted rules quantitatively (Figure 2): mean predicted ΔMRL of -0.43 / -1.38 / -2.39 for in-frame / uORF / overlapping-ORF insertions, against ground truth -0.40 / -1.36 / -2.52 ($r = 0.95$ across all insertions). The model has learned the frame logic, not merely "ATG is bad".

On the real data, training from scratch with the published recipe gives $r^2 = 0.919 \pm 0.006$ over three seeds (0.924, 0.921, 0.910), against the published 0.93 (Figure 1). The residual gap is consistent with seed variance: with only three

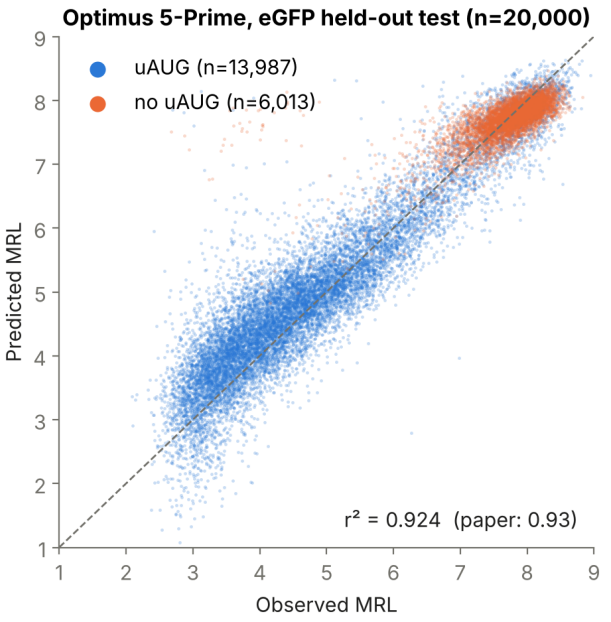


Figure 1. Reproduction of the paper's Figure 2b. Predicted vs. observed MRL on the 20,000 held-out UTRs, coloured by presence of an upstream AUG. Our from-scratch model reaches $r^2 = 0.924$ on this seed (published: 0.93). The uAUG-free sequences (orange) occupy the high-MRL mode, as expected biologically.

epochs, no learning-rate schedule and no early stopping, run-to-run spread is ≈ 0.01, and the paper reports a single run. Under our proper train/validation/test protocol with early stopping the same architecture gives $r^2 = 0.922 \pm 0.002$.

Biological sanity checks. Independently of the model, the data reproduce the published frame effect: among single-uAUG UTRs, out-of-frame uAUGs depress MRL to ≈ 4.5–5.5 while in-frame uAUGs are far milder and approach uAUG-free levels close to the CDS. We also reproduce the human-UTR evaluation (Figure 4a of the paper) at $r^2 = 0.784$; the published 0.81 uses a *retrained* model whose weights and exact recipe are not in the authors' repository. The authors' own original model scores 0.789 on our pipeline, so the gap is attributable to that missing model rather than to our implementation.

5. Benchmark Comparison

We compare Optimus 5-Prime against three benchmarks, all trained and evaluated under the identical protocol of Section 3:

- **Position-specific k-mer linear model** ($k = 1 \ldots 6$, ridge regression) — the paper's own baseline;
- **LightGBM** (Ke et al., 2017) on hand-crafted features: uAUG counts by functional type, best Kozak context per type, stop-codon counts per frame, distance of first/last uAUG, GC content, MFE, and 1–3-mer counts (98 features);
- **BiLSTM** (Hochreiter and Schmidhuber, 1997) (bidirectional LSTM, mean+max pooling) — a different deep-learning family on the same one-hot input.

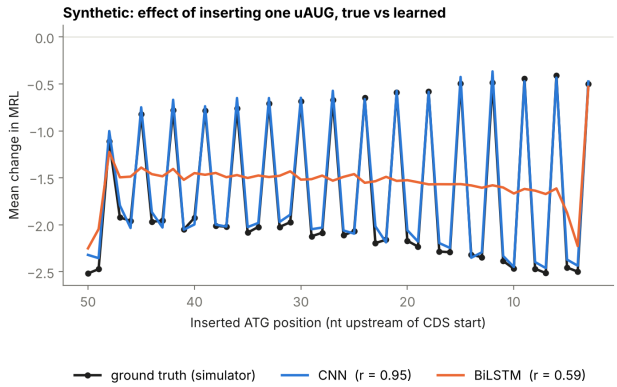


Figure 2. Rule recovery on synthetic data. A single ATG is inserted at every position of 400 uAUG-free test UTRs and the predicted change in MRL is compared with the simulator's true change. The CNN tracks the ground truth, including the three-nucleotide frame periodicity; the BiLSTM learns only that ATG is repressive.

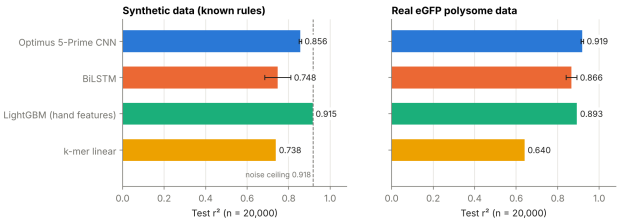


Figure 3. Benchmark comparison on synthetic data (left, with computable noise ceiling) and real eGFP data (right). LightGBM's synthetic score is near-oracle because its hand-crafted features encode the very rules the simulator plants.

Because the task is regression but the rubric calls for classification metrics, we additionally bin MRL into **tertiles of the training distribution** (low $< 5.98 \leq$ medium $< 7.63 \leq$ high) and report accuracy and macro-averaged precision, recall and F1, with confusion matrices. Table 1 gives the unified comparison on real data and Figure 4 the confusion matrices; Figure 3 contrasts synthetic and real performance.

Interpretation. Three observations matter more than the ranking itself.

(i) The deep model earns its place, but not by as much as the paper's baseline suggests. The published comparison is against a k-mer linear model (0.66 published, 0.640 reproduced here). A properly tuned gradient-boosted model on biologically informed features reaches 0.893 — far closer

Table 1. Unified comparison on the real eGFP held-out test set ($n = 20{,}000$). Rows above the rule are the reproduction and the three benchmarks; rows below are the architectures compared in Section 6.1. Classification metrics use training-set tertiles of MRL; precision, recall and F1 are macro-averaged. Mean ± sd over seeds where more than one seed was run.

Model	Role	Seeds	r^2	Spearman	RMSE	Accuracy	Precision	Recall	F1
Optimus 5-Prime CNN	reproduction	3	0.919±.006	0.942	0.547	0.869	0.822	0.822	0.819
k-mer linear	benchmark	1	0.640	0.832	1.155	0.696	0.723	0.659	0.636
LightGBM (hand features)	benchmark	1	0.893	0.922	0.620	0.848	0.795	0.791	0.791
BiLSTM	benchmark	3	0.866±.026	0.907	0.746	0.819	0.773	0.769	0.763
Transformer	ours	3	0.728±.012	0.835	1.013	0.713	0.704	0.655	0.635
CNN + attention hybrid	ours	3	**0.942**±.001	**0.955**	**0.454**	**0.886**	**0.839**	**0.843**	**0.841**

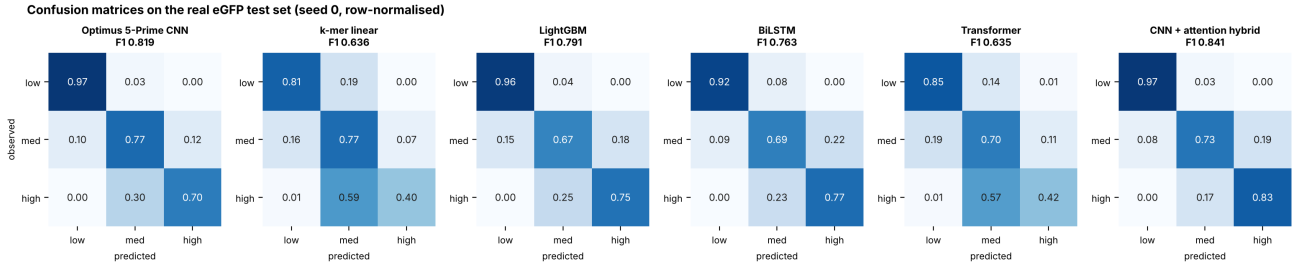


Figure 4. Row-normalised confusion matrices on the real test set (seed 0) for MRL binned into training-set tertiles. Every model finds the low-MRL class easiest, because a single uAUG is a strong and easily learned signal; the high class is hardest, requiring the absence of repressive elements *and* favourable structure. The CNN+attention hybrid (rightmost) is the only model whose recall on the high-MRL class exceeds 0.80.

to the CNN. The honest claim is that the CNN beats strong feature engineering by $\approx 0.03\, r^2$ while requiring no domain features at all.

(ii) LightGBM's synthetic score is near-oracle and should not be read as a win. On synthetic data it scores 0.915 against the ceiling of 0.918, but its features (uAUG type, Kozak strength, MFE) are precisely the quantities the simulator uses. This is exactly the diagnostic value of the synthetic set: it separates "learned the task" from "was handed the answer".

(iii) Error structure is biologically sensible. All confusion matrices are strongly diagonal with errors confined to adjacent classes; no model confuses low with high MRL. The low class is recovered best (0.92–0.97 recall for the stronger models), consistent with a single out-of-frame uAUG being a near-deterministic repressive signal.

6. The CNN–Attention Hybrid

The reproduction shows that the published CNN is a strong model, and the benchmark comparison narrows its margin over well-tuned feature engineering to $\approx 0.03\, r^2$ (Section 5). Neither result tells us whether the convolutional inductive bias is the *right* one for this task or merely a sufficient one, because the original work evaluates a single architecture at a single data size. There is a concrete reason to ask. A convolutional stack reads a fixed receptive field and can only compose distant positions through depth, whereas the biology is explicitly about interaction between distant positions: how much a given uAUG represses depends on its Kozak context, on whether a stop codon occurs later in frame, and on structure elsewhere in the 50 nt window. Self-attention models such interactions directly, which raises two questions — whether attention helps at all here, and whether it helps on its own or only on top of a convolutional stem.

We therefore run one controlled architecture × data-regime study, and it yields the single extension we propose: a hybrid that feeds a convolutional stem into a transformer encoder. Everything else about the setup — the data, the splits, the test set and the evaluation protocol — is unchanged from Section 3, so the comparison against the reproduced CNN is like-for-like.

6.1. Architectures across data regimes

The paper evaluates one architecture at one data size. We compare three at $\{1\text{k}, 5\text{k}, 260\text{k}\}$ training sequences: the CNN (475k parameters), a pre-LN transformer encoder (Vaswani et al., 2017) over the 50 positions with learned positional embeddings and mean pooling (457k), and a hybrid of two convolutional layers feeding a three-layer transformer (421k). Fairness controls: matched parameter counts, an identical tuning grid per architecture per data size (learning rate × dropout, selected on validation), the same early-stopping rule, five seeds at 1k/5k and three

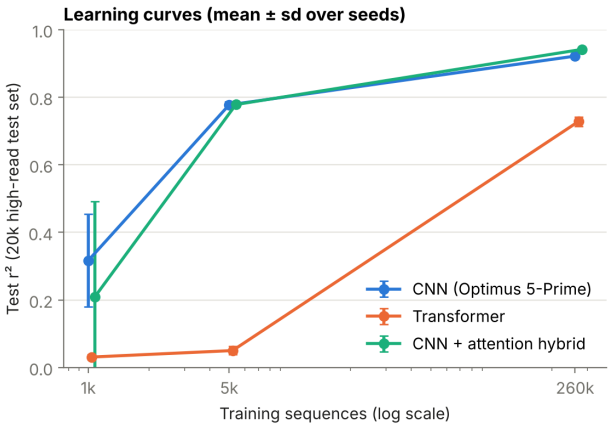


Figure 5. Learning curves. Parameter counts are matched (475k / 457k / 421k) and every architecture receives the same tuning budget, early-stopping rule and seeds at each data size. The transformer's full-data point is budget-limited (see text).

Table 2. Test r^2 by architecture and training-set size (mean ± sd; five seeds at 1k/5k, three at 260k). †All three full-data transformer runs hit the 15-epoch cap while validation loss was still falling, so this value is budget-limited rather than a converged ceiling.

Train size	CNN	Transformer	Hybrid
1,000	**0.317**±.123	0.031±.007	0.210±.251
5,000	**0.778**±.009	0.050±.011	**0.778**±.005
260,000	0.922±.002	0.728†±.012	**0.942**±.001

at full data, and nested random subsets drawn only from the training pool.

Table 2 and Figure 5 give the result. **The hybrid is the best model at full data**, reaching $r^2 = 0.942 \pm 0.001$ — above the published CNN on all three seeds, with fewer parameters and lower RMSE (0.454 vs. 0.522). The convolutional stem supplies the local motif prior that attention otherwise has to discover, while self-attention adds the long-range composition the CNN lacks.

The pure transformer fails, and the failure is informative. At 1k and 5k it cannot fit even its training set. We verified this is not an implementation bug using training data only: changing the learning rate does not help, replacing the learned positional embedding with a sinusoidal one does not help, and the identical encoder behind a convolutional stem (the hybrid) trains normally. At full data it sits on a plateau for roughly eight epochs (≈ 14k updates) before breaking out, and all three runs were stopped by the shared 15-epoch cap while still improving. Learning a motif such as ATG from single-nucleotide tokens through attention alone requires far more optimisation steps than the convolutional prior needs.

Sensitivity and controls. Because the hybrid is bimodal at 1k (either ≈ 0.65 or ≈ 0.04, depending on whether it escapes the initial plateau), we repeated the small-data runs with patience 30 instead of 10 for *all* architectures. This lifts the CNN at 1k from 0.317 to 0.435 ± 0.054 — the shared rule was under-training the CNN too — but does not rescue the transformer (0.035) nor remove the hybrid's bimodality, and the ranking at 5k is unchanged. On a random (rather than read-depth) test split, single-seed full-data scores are CNN 0.891 and hybrid 0.896 versus 0.923 and 0.941 on the high-read split: the modest drop is expected since random test sets include noisier low-coverage UTRs, and the ordering holds.

6.2. Attribution agreement with the CNN

A model that scores higher could be doing so by exploiting different, and possibly spurious, features. To check, we ran exhaustive single-nucleotide in-silico mutagenesis (ISM) on 2,000 held-out test UTRs for the CNN, the hybrid and the pure transformer: every substitution at every position is scored, and the resulting attribution maps are compared per sequence.

The hybrid's ISM maps agree with the CNN's at per-sequence $r = 0.96$, with 84% overlap in the top-5 attributed positions — nearly as close as two CNN seeds agree with each other ($r = 0.97$). The hybrid therefore improves by *refining* a grammar the CNN already learns, not by learning a different one; the gain is not an artefact of the model attending to something the assay does not support. The pure transformer agrees considerably less ($r = 0.79$, 64% overlap) and relies less on uAUGs, which is consistent with its weaker scores.

7. Discussion and Limitations

Our positive result is the hybrid architecture: $+0.020$ r^2 over the published CNN under an identical protocol, consistent across all three seeds, with *fewer* parameters (421k vs. 475k) and lower RMSE (0.454 vs. 0.522). It also survives the check that matters most for a model meant to be used rather than only scored: its attribution maps track the CNN's almost as closely as two CNN seeds track each other, so the improvement refines a grammar we can already interpret. For mRNA design, where such a model is used to rank candidate 5′UTRs, the margin comes at no cost in domain assumptions — the hybrid takes the same one-hot 50 nt input as the original.

Several limitations should be stated plainly. **Compute.** All experiments ran on two CPU cores without a GPU, which forced a 15-epoch cap at full data. The transformer in particular was still improving when stopped, so its ceiling is unknown, and the hybrid's margin over the CNN is estab-

lished only within that shared budget. **Variance at small data.** At 1k the hybrid is bimodal and standard deviations are large; conclusions there are about typical-case behaviour under a fixed budget, not about asymptotic capacity. **Scope.** Every experiment reported here uses the unmodified (U) eGFP library and the fixed 50 nt input of the original assay. We make no claim about the modified-nucleotide (Ψ, m1Ψ) chemistries, about variable-length 5′UTRs, or about variant-effect prediction; extending the hybrid to those settings is left to future work.

8. Conclusion

We reproduced Optimus 5-Prime faithfully — $r^2 = 0.919 \pm 0.006$ from scratch against a published 0.93, and 0.934 when the authors' own released weights are loaded into our implementation — after first validating the pipeline on a synthetic dataset whose rules we control, and we benchmarked the model against three alternatives under one protocol with both regression and classification metrics.

We then proposed one focused extension. A controlled architecture $\times$ data-regime study, with matched parameter counts and matched tuning budgets, shows that a hybrid of a convolutional stem and a transformer encoder reaches $r^2 = 0.942 \pm 0.001$ and is the strongest model in our comparison, while a pure transformer on the same input fails in low-data regimes. Read together, these say that the convolutional prior is necessary but not sufficient: attention is worth adding *on top of* it rather than in place of it. In-silico mutagenesis confirms that the hybrid refines the grammar the CNN already learns rather than substituting a different one. The margin is established within a fixed 15-epoch CPU budget, and confirming it under a larger budget is the obvious next step.

Code and Reproducibility

All code, trained weights, per-run metrics (including validation curves and seeds), saved subset indices, and the scripts that regenerate every figure and table in this report are at https://github.com/anikag06/CHE_629.

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